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Dynamic Calibration of Decision Thresholds for Financial Anomaly Detection:

Verification With Payment Platform Information and Data

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ABSTRACT

Digital payment channels have expanded quickly, reshaping transaction flows while opening new avenues for fraud. Isolation Forest (IF) remains attractive for unsupervised screening, yet deployments that rely on a fixed anomaly-score threshold deteriorate when traffic shifts or is actively manipulated. The authors present a Temporal-Attention Isolation Forest with Dynamic Calibration (TA-IFDC) that treats threshold selection as an adaptive component rather than a static post-processing step. The method monitors the evolving distribution of IF scores in streaming mode and updates the decision boundary online, while a lightweight temporal-attention module encodes short-range dependencies across consecutive transactions. Together, these pieces allow the detector to adjust to drift without sacrificing precision during stable periods.

KEYWORDS

Financial Transaction Security, Isolation Forest, Adaptive Decision Threshold, Temporal Dependency Modeling, Real-Time Fraud Analytics

1. INTRODUCTION

Over the last decade, the accelerating expansion of digital financial services has altered the fundamental architecture of global payment systems. Mobile banking applications, instant online transfers, and integrated e-wallet ecosystems have rapidly displaced traditional cash and card-based

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transactions, creating a financial environment that is faster, more accessible, and highly interconnected. This transformation has yielded substantial benefits in terms of operational efficiency and user convenience; however, it has also brought about a new spectrum of security vulnerabilities that financial institutions must confront. Recent statistical surveys place the total global losses attributed to payment fraud at more than USD 45 billion in 2022, with forecasts indicating a continued upward trajectory as illicit actors incorporate automation, large-scale botnets, and adversarial machine learning into their operations. Market analyses further note that fraud patterns are becoming increasingly fragmented and adaptive, making the early detection of suspicious transactions a more complex task than ever before (Zhang et al., 2022).

The emergence of such high-frequency and high-value fraudulent transactions underscores the necessity for monitoring systems capable of functioning effectively under volatile, non-stationary data conditions. Modern fraud detection commonly relies on anomaly detection frameworks that assess deviations from established transactional norms(Hernandez Aros et al., 2024). Unsupervised detectors based on IF are widely used because they run fast on large datasets and do not rely on labels (Janjua et al., 2024; Kareem & Muhammed, 2024). IF builds random partition trees by choosing attributes and split values at random. Points that depart from the bulk are isolated in only a few splits, so their path lengths are short. In practice, this behavior makes IF a good fit for catching rare but consequential events in high-dimensional data without costly annotation.

Experience with IF in production exposes a weak spot: decisions usually depend on a fixed score cutoff set during an initial calibration or chosen heuristically, then left in place. Real payment traffic does not sit still—seasonal campaigns, macro shifts, and coordinated rings all move the score distribution (Sonani & Govindarajan, 2022). A static line can flood reviewers during benign surges or miss gradual shifts that matter; both outcomes degrade performance.

Streaming tightens the constraints. Each transaction must be scored in milliseconds, while labels often arrive late or not at all(Vanini et al., 2023). The detector acts under uncertainty as the score distribution drifts, and a fixed threshold cannot follow that movement. Deeper models and feature tweaks help, but a robust, feedback-aware way to adjust cutoffs in unsupervised settings is still underexplored.

To address this, we introduce TA-IFDC and treat thresholding and temporal context as core parts of the detector (Attar et al., 2024). A dynamic calibration step updates the decision boundary online using recent score statistics and delayed outcomes, keeping alert rates stable while tracking distribution change (Al Lawati et al., 2025; Lin et al., 2024). A lightweight temporal component encodes short-range dependencies—inter-arrival times and simple session cues—so scores reflect what just happened rather than judging each record in isolation (Zheng et al., 2025).

We evaluate the framework on five payment datasets—IEEE-CIS, PaySim, CCFD, SFD-FD, and BankSim—against six competitive baselines. We report precision, recall, F1, AUC, and per-transaction latency under streaming protocols that preserve order, inject drift, and delay labels to mirror practical deployment.

Together, adaptive thresholding and temporal context close the gap between raw anomaly scores and real-time decisions in payment systems, yielding consistent gains while meeting the latency constraints of high-throughput gateways.

By coupling adaptive thresholding with temporal sequence modeling, TA-IFDC bridges a critical gap between raw anomaly scoring and real-time decision-making in financial fraud detection systems. (Tchuente, 2022) The results not only demonstrate the framework's ability to deliver consistent performance gains across multiple datasets but also highlight its suitability for deployment in high-throughput payment platforms where both fraud patterns and data distributions can change rapidly. This combination of operational adaptability and computational efficiency positions TA-IFDC as a promising candidate for next-generation fraud detection infrastructure in the financial sector (Bello et al., 2024; Fatlawi, 2025).

The primary contributions of this work are summarized as follows:

(1) A dynamic threshold calibration mechanism is proposed, enabling IF to operate adaptively in non-stationary data environments without external supervision;
(2) A temporal attention module is introduced to model inter-transaction dependencies, improving detection accuracy for sequential or contextual anomalies(Tokovarov & Karczmarek, 2022);
(3) A unified detection framework—TA-IFDC—is constructed and evaluated across multiple real and synthetic payment datasets, demonstrating consistent improvements in accuracy, robustness, and latency over baseline models;
(4) Extensive experiments and ablation studies provide insight into the behavior of dynamic calibration mechanisms and the value of temporal modeling under streaming and drifting conditions.

The remainder of this paper is structured as follows. Section 2 presents a review of related work in fraud detection, threshold calibration, and time-aware anomaly modeling. Section 3 describes the architecture and core components of the proposed TA-IFDC framework. Section 4 details the experimental setup, including datasets, evaluation metrics, and baseline configurations. Section 5 presents and interprets the experimental results. Section 6 discusses the research findings, limitations, and implications. Finally, Section 7 concludes the paper and outlines directions for future work.

2. RELATED WORK

The increasing reliance on unsupervised anomaly detection techniques in the financial domain has drawn extensive attention to IF and its variants. Originally proposed by Liu et al., the IF algorithm isolates observations by randomly selecting features and split values, under the assumption that anomalies are more susceptible to early isolation(Al Farizi et al., 2021; Hilal et al., 2022; Immadisetty, 2025). Its efficiency and effectiveness have rendered it a foundational tool in large-scale and high-dimensional fraud detection tasks. However, the original IF model lacks adaptability to time-varying data patterns and fails to incorporate contextual or sequential transaction information, which are prevalent in financial fraud scenarios(Ali et al., 2022; Du & Shu, 2022; Kamuangu, 2024; Quan et al., 2024).

Efforts to overcome the inherent limitations of the IF algorithm have taken many forms, though the core motivation remains the same: improve adaptability without sacrificing interpretability(Lam, 2025; Shanaa & Abdallah, 2025; Wang et al., 2023). Early adaptations for streaming data, typified by the Online-iForest approach, allowed incremental updates to the ensemble of trees, enabling the model to handle new transactions as they arrive(Immadisetty, 2025; Leveni et al., 2025). This shift from batch processing to a more fluid, real-time structure was a practical leap for high-frequency domains such as payment platforms. Other lines of work explored hybrid architectures that combine IF with ensemble classifiers or incorporate domain-specific engineered features to boost discrimination accuracy, particularly in noisy, high-dimensional datasets(Koziara & Karczmarek, 2022; Núñez Delafuente et al., 2024; L. Zhang et al., 2025). More recently, graph-based formulations—exemplified by GNN-IF—embed transaction records into account–entity graphs, capturing relational signals that tree models in isolation would overlook(Chen & Tsourakakis, 2022; Kim et al., 2022). While these methods offer structural flexibility or richer representation learning, they still inherit the same fixed thresholding mechanism as the vanilla IF, which inevitably curtails responsiveness when fraud patterns shift unexpectedly. This is a notable bottleneck, especially in adversarial settings where attackers deliberately exploit static detection boundaries.

Calibrating decision thresholds remains underexplored relative to model design in anomaly detection. In many mature pipelines, the anomaly score is mapped to a decision through a single operating point chosen at launch and rarely revisited. Under non-stationary traffic, that practice is brittle: concept drift, class-prior shift, and seasonal volume swings push the score distribution away from the original calibration, tilting the false-positive/false-negative mix. In financial settings the asymmetry is costly—missed fraud yields direct loss, while excessive alerts erode reviewer trust and inflate operational budgets. Recent work on adaptive thresholding has opened up useful directions, particularly in situations where labels arrive late or sparsely, and batch evaluation isn't feasible. Many systems rely on simple rules—rolling quantiles, for example—to keep the decision boundary in sync with shifting score distributions, while others incorporate feedback from downstream performance, nudging thresholds over time to maintain a reasonable precision–recall balance. There's also a stream of research that treats thresholding as a policy problem, often applying reinforcement learning to track evolving attacker behavior. However, in practice, two issues tend to surface. First, when the data distribution is heavy-tailed or the traffic arrives in bursts, naïve update windows can behave erratically, producing unstable alert patterns. Second, most of these approaches still assume access to trustworthy labels, which are often unreliable or delayed in real-world environments. Tools like Platt scaling or isotonic regression help calibrate probabilities, but they don't solve the problem of where to actually place the operating point. So while adaptive strategies show potential, especially in dynamic or semi-supervised settings, their practical value is still constrained by label availability. In these cases, approaches that combine anomaly scoring with lightweight, feedback-tolerant calibration—without relying heavily on labels—remain highly relevant. A related line of work frames threshold control as a learning policy, often via reinforcement learning (RL). Such methods tune a reward reflecting precision–recall trade-offs and adapt under drift, but they depend on frequent, high-quality feedback—rare in production payment streams. TA-IFDC instead calibrates from score distributions and sparse, delayed outcomes, achieving responsiveness without a heavy supervision loop.

In practical financial environments, anomaly detection systems are required to cope with a wide spectrum of transaction types, the continual emergence of new fraud strategies, and the inherent variability of streaming data(Eswar Prasad et al., 2023; H. Zhang et al., 2025). Scholars have examined a range of detection paradigms, from supervised models—such as gradient-boosted decision trees (e.g., XGBoost), deep architectures like autoencoders, to hybrid frameworks that merge clustering outputs with classification layers—to address these challenges(Almazroi & Ayub, 2023; Mazumder et al., 2025). Although supervised learning methods can deliver strong accuracy when abundant annotated data are available, their dependence on up-to-date labels often limits their flexibility and long-term maintainability once deployed. On the other hand, unsupervised approaches, including Isolation Forest, one-class Support Vector Machines (SVM), and various clustering schemes, avoid the label dependency and thus can be applied more broadly across heterogeneous datasets. Nevertheless, these methods frequently encounter difficulties in providing transparent decision logic and in setting decision thresholds that remain effective under shifting data distributions. The datasets most commonly used for evaluation include the IEEE-CIS Fraud Detection Dataset, PaySim, the CCFD dataset, and BankSim. Each dataset presents unique characteristics—ranging from anonymized features and class imbalance to temporal drift and behavior simulation—requiring detection frameworks to be robust across multiple conditions.

Compared with existing approaches, the method proposed in this paper distinguishes itself through its integration of dynamic threshold calibration with an unsupervised anomaly detection backbone. Rather than designing new feature representations or model architectures, this study focuses on the often-overlooked thresholding component and enhances it with a feedback-informed and context-aware strategy. The TA-IFDC framework combines a temporal attention mechanism with an adaptive calibration module, allowing the model to respond dynamically to score fluctuations and transaction context. This design improves anomaly interpretability, reduces false alarms, and preserves the unsupervised nature of IF-based models, making it more suitable for real-world deployment where labeled data are sparse and latency constraints are strict.

In summary, while previous studies have improved various facets of anomaly detection models—such as feature representation, network architecture, or data stream handling—few have systematically addressed the calibration challenge within unsupervised fraud detection frameworks. This study aims to fill that gap by introducing a threshold adaptation mechanism that complements the strengths of IF, while enhancing temporal responsiveness and practical applicability.

3. PROPOSED METHOD: TA-IFDC

We propose a novel anomaly detection framework named TA-IFDC, designed to enhance the performance of IF through adaptive threshold adjustment and temporal-context modeling. This section details the system architecture, the role of each component, and the mathematical formulation of the algorithm.

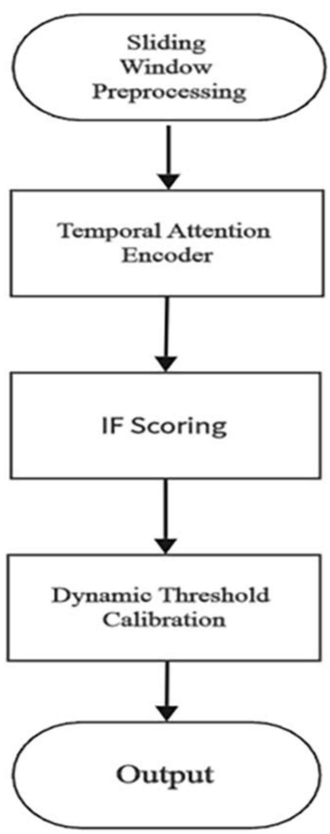
3.1 Overview of TA-IFDC

The TA-IFDC framework is designed with four major components:

1. Sliding Window Preprocessing for temporal transaction segmentation;
2. Temporal Attention Encoder, which captures transaction time patterns and user behavior drift;
3. IF Scoring, responsible for initial anomaly detection;
4. Dynamic Threshold Calibration Module, which adaptively adjusts the decision boundary based on recent detection performance.

A high-level schematic of the method is shown in Fig 1 .

Figure 1. Overview of the main processing stages in TA-IFDC



3.2 Sliding Window-Based Transaction Segmentation

To enable temporal modeling and online threshold calibration, transaction data is divided into overlapping windows.

Let the incoming transaction stream be denoted as:

$$\mathscr{T} = \{x_1, x_2, \ldots, x_N\}, x_i \in \mathbb{R}^d$$

where $\mathbf{x}_i$ represents a single transaction with d features.

We construct a sliding window of size W, with stride S, resulting in a sequence of windows:

$$\mathscr{W}_k = \{x_{kS}, \ldots, x_{kS+W-1}\}, k = 1, \ldots, K$$

This segmentation supports online model updates and threshold adaptation.

3.3 Temporal Attention Encoder

To capture periodic patterns and behavior drift, we embed a temporal attention mechanism that computes the temporal relevance of each transaction within a window.

Let:

- $X_k = [x_{k,1}, \ldots, x_{k,W}]$ denote the feature matrix of window $\mathscr{W}_k$
- $T_k = [t_{k,1}, \ldots, t_{k,W}]$ denote the corresponding time-stamps

We define time encoding vectors $e_{k,j} = \phi(t_{k,j}) \in \mathbb{R}^h$, where $\phi(\bullet)$can be a Fourier-based positional encoding or learned embedding.

Each transaction is transformed into:

$$\tilde{x}_{k,j} = concat(x_{k,j}, e_{k,j})$$

We then compute attention scores using scaled dot-product attention:

$$\alpha_{ij} = \frac{\exp(q_i^\top k_j / \sqrt{d_k})}{\sum_{l=1}^{W} \exp(q_i^\top k_l / \sqrt{d_k})}$$

where:

- $q_i = W_Q \tilde{x}_{k,i}$ is the query vector
- $k_j = W_K \tilde{x}_{k,j}$ is the key vector
- d_k is the dimension of keys

The attention-enhanced transaction vector becomes:

$$z_i = \sum_{j=1}^{W} \alpha_{ij} \cdot (W_V \tilde{x}_{k,j})$$

These contextual representations z_i are then passed to the IF model for scoring.

3.4 IF Scoring

The enhanced representations z_i are input into the IF to calculate anomaly scores.

IF relies on recursive partitioning. For each instance z_i, the anomaly score $s(z_i)$ is computed as:

$$s(z_i) = 2^{-\frac{E(h(z_i))}{c(n)}}$$

where:

- $E(h(z_i \backslash))$ is the average path length of z_i in all trees
- n is the subsample size used to build trees
- c(n) is the normalization factor:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

and H(i) is the harmonic number:

$$H(i) = \ln(i) + \gamma, \gamma \approx 0.5772$$

A score close to 1 indicates high anomaly probability.

3.5 Dynamic Threshold Calibration

In traditional IF implementations, the decision threshold θ\theta is fixed, often determined by cross-validation or empirical heuristics. However, under real-world financial settings, user behavior patterns and transaction distributions can drift significantly over time, resulting in unstable detection performance if the threshold remains static.

To address this, we design a Dynamic Threshold Calibration (DTC) module that adaptively adjusts θ\theta for each sliding window using recent prediction feedback statistics.

3.5.1 Adaptive Threshold Update Rule

Let $\mathcal{S}_k = \{s_1, s_2, \ldots, s_W\}$ be the anomaly scores in window $\mathcal{W}_k$, and let θ_k be the calibrated threshold at time step k. We define a percentile-based update rule with smoothing:

$$\theta_k = (1-\lambda) \cdot \theta_{k-1} + \lambda \cdot Quantil\, e_\beta(\mathcal{S}_k)$$

where:

- $\lambda \in [0,1]$ is the learning rate (adaptivity coefficient),
- $\text{Quantile}_\beta(\bullet)$returns the β-th percentile of the scores (e.g., β=0.95 for top 5% anomaly),
- θ_0 is initialized from the first window as baseline.

This rule smooths threshold changes to prevent instability due to local score fluctuations.

3.5.2 Feedback-Informed Calibration (Optional Enhancement)

When delayed ground truth labels are available (e.g., confirmed fraud), a feedback loop is used to refine θ_k based on false positive and false negative counts.

Let:

- FP_k: false positives in $\mathscr{W}_k$,
- FN_k: false negatives,
- $\Delta\theta_k$: correction factor.

We define:

$$\Delta\theta_k = \eta \cdot \left(\alpha \cdot \frac{FP_k}{W} - (1-\alpha) \cdot \frac{FN_k}{W}\right)$$

and update the threshold as: $\theta_k \leftarrow \theta_k + \Delta\theta_k$ where:

- η is the feedback gain (tuning parameter),
- $\alpha \in [0,1]$ balances precision and recall.

This formulation penalizes overly aggressive thresholds (high FP) and overly conservative thresholds (high FN), encouraging balance.

3.6 Overall Algorithm Description

The TA-IFDC framework is designed to perform real-time anomaly detection in financial transaction streams, with particular emphasis on adaptability to evolving data distributions. It integrates temporal behavior modeling, ensemble anomaly scoring, and an adaptive decision-making mechanism to accommodate the non-stationary nature of payment systems.

Initially, the transaction stream is segmented using a sliding window approach. Each window contains a sequence of temporally ordered transaction records, which serves as the processing unit for detection. This segmentation enables localized modeling of user behavior and supports gradual updates of the decision threshold over time.

For each window, temporal characteristics of transactions are encoded into fixed-length feature representations. These time-sensitive features are derived using positional encoding techniques and capture both the periodicity and recency of financial activities. To further enhance context-awareness, a temporal attention mechanism is applied. The mechanism assigns varying degrees of importance to transactions observed within a given time window, enabling the model to focus more on events that carry greater behavioral significance, while downplaying patterns that contribute little to the detection objective.

After temporal encoding, each transaction is processed through an ensemble of Isolation Trees, which partition the feature space in a randomized manner. IF does not chase balanced splits. It samples a feature and a cut at random at each node, and the isolation depth depends on how far the transaction sits from the main mass. Transactions that deviate substantially from the bulk of data are separated earlier, producing shorter path lengths and correspondingly higher anomaly scores. In financial datasets, such cases often represent unusual spending bursts, irregular transfer patterns, or atypical device usage, all of which warrant closer scrutiny. The unsupervised nature of this scoring process is particularly important in fraud detection, where labeled anomalies are scarce, delayed, or incomplete. Moreover, the low computational cost of Isolation Trees allows the method to scale to millions of transactions per day without prohibitive infrastructure requirements.

Once the anomaly scores for a given observation window are available, they serve as the basis for recalibrating the detection threshold. The TA-IFDC framework avoids the rigidity of static thresholds by adopting a quantile-based dynamic calibration approach. In practical terms, this means that the decision boundary at time t is updated by blending the previously applied threshold with a high-percentile statistic—such as the 95th or 97th percentile—of the current score distribution. The blend ratio controls how quickly the system reacts to abrupt changes, such as a sudden spike in fraudulent activity following a phishing campaign. A purely percentile-based recalibration could overreact to noise, while a purely historical threshold would be slow to adapt; the interpolation mechanism balances these extremes, maintaining stability without sacrificing responsiveness. Quantile selection was ablated across 90–99 percentiles. Below ~93% false positives surged; above ~97% recall fell sharply. We adopt 95% as a balanced operating point, with $\pm$1–2% guardrails under drift. To avoid overreaction, a hysteresis rule and adaptive smoothing cap limit per-window threshold change to $\approx$2–3% of the prior value.

In operational settings, it is common for feedback on suspected transactions to arrive only after several hours or even days, once investigations are complete. TA-IFDC is designed to incorporate such delayed supervision when available. Specifically, the recalibration step introduces a correction term proportional to the recent imbalance between false positives and false negatives. When false positives rise, the threshold is nudged upward to protect analyst capacity. If post-review reveals missed fraud, the threshold is lowered to recover recall. Updates use delayed labels when available and otherwise rely on recent score dynamics, allowing the operating point to adapt under partial supervision typical of large-scale payment streams.

After each threshold update, transactions in the active window are scored against the current operating point. Records above the cutoff move to the risk controls. Depending on policy and traffic, these controls may act automatically—short holds, step-up authentication, velocity caps—or hand the case to investigators. TA-IFDC runs in mini-batch mode and in an online (prequential) loop, while meeting real-time and accuracy requirements. When a window finishes, the pointer slides forward and calibration continues on the next slice, letting the system keep pace with the stream.

Temporal feature encoding, context-aware attention, isolation-based scoring, and online threshold calibration jointly counter concept drift and ongoing adversarial changes. Legitimate behavior swings with sales, public events, and product launches; meanwhile, attackers keep changing tactics to slip past static rules. By making calibration part of the design rather than a post-hoc tweak, TA-IFDC ties score production to the decision rule and closes the long-standing gap between scoring and action. The modules are small and loosely coupled, making integration with existing case-management and risk-control systems straightforward.

We evaluate on five benchmarks—IEEE-CIS Fraud Detection, PaySim, CCFD, SFD-FD, and BankSim—spanning real and simulated payment streams with varied class priors and feature sets. We assess precision, recall, F1, AUC, per-record latency, and robustness under induced drift and delayed feedback under constraints similar to production.

4. EXPERIMENTAL SETUP

We evaluate TA-IFDC under conditions close to our target deployment. The focus is on behaviors that matter in practice. We use five benchmarks—IEEE-CIS, PaySim, CCFD, SFD-FD, and BankSim—to cover large-scale online payments and simulated account-takeover cases. Using both real and synthetic data lets us probe stationary regimes as well as non-stationary patterns such as seasonal shifts and tactic drift.

For comparability, one protocol is applied across datasets. Each dataset is ordered chronologically and split 80%/10%/10% into train/validation/test by time, preventing look-ahead leakage—crucial for streaming or time-dependent transactions. We report precision and recall for the false-alarm/miss trade-off, F1 as their harmonic summary, AUC for ranking across thresholds, and per-record latency for near-real-time constraints. As a set, these metrics give a more faithful picture than accuracy on heavily imbalanced data.

We benchmark against six baselines: Online-iForest, Hybrid-AI, GNN-IF, XGB-Anomaly, SSR-RVFL, and an autoencoder-based detector. Each baseline follows its paper and public code when available; otherwise, hyperparameters are tuned on the validation split. This keeps implementations faithful to their intended design, so any gaps reflect modeling choices rather than configuration quirks.

4.1 Datasets

For the empirical evaluation of the TA-IFDC framework, we selected five representative financial transaction datasets: IEEE-CIS Fraud Detection, PaySim, CCFD, SFD-FD, and BankSim. Together, these datasets encompass a wide spectrum of operational contexts, ranging from large-scale real-world payment records to semi-synthetic and fully simulated banking scenarios. They not only reflect static historical fraud patterns but also capture environments where fraudulent behavior evolves over time, sometimes in response to seasonal trends or system countermeasures. The mix of real transactions and synthetic traces exposes the method to distribution shift, class-prior variation, and non-stationary temporal patterns, offering a clearer view of detection performance and adaptability.

The IEEE-CIS Fraud Detection dataset (originally released for the 2019 IEEE-CIS Fraud Detection challenge on Kaggle) contains more than one million labeled online transactions. The dataset includes device and browser fingerprints, timestamps, and a mix of anonymized fields describing user behavior and payment context. The overall fraud rate sits around 3.5%—roughly one in twenty-eight cases—which is close to what's seen in production systems. Because the features are high-dimensional and partly anonymized, it's a good test bed for anomaly detectors that need to catch rare events without overreacting to noise. The time-stamped nature of the records also exposes gradual month-to-month drifts in device types and behavior, which made it useful for testing how the temporal-attention block and the dynamic threshold module in TA-IFDC react to changing conditions. For comparison, we also used PaySim, a synthetic mobile-money dataset built from agent-based simulations of real transaction behavior observed in sub-Saharan Africa. It comprises over 6 million transactions across five operational categories, including "TRANSFER" and "CASH-OUT". Although synthetic, PaySim mirrors real user interaction patterns, offering a controlled yet realistic testing ground. The inherent low fraud prevalence (approximately 0.13%) and the ease of injecting artificial concept drift make it ideal for testing the model's responsiveness to behavioral shifts.

The CCFD dataset, provided by a European card issuer, contains approximately 285,000 credit card transactions, with features anonymized using principal component analysis. Fraudulent cases represent a mere 0.172% of the data, making it an extreme example of class imbalance. While the feature structure is static, it remains a widely accepted benchmark for validating anomaly detection models, particularly with respect to precision, recall, and F1-score.

The SFD-FD dataset, published by IBM, is a synthetically generated dataset rooted in real fraud patterns observed in banking and financial services. It includes temporal transactions, labeled fraud cases, and control variables designed to simulate concept drift scenarios, such as seasonal changes or policy shifts. This dataset offers controllable drift events, which is essential for validating the adaptability of the dynamic threshold mechanism under non-stationary data distributions.

BankSim is a synthetic corpus generated with a multi-agent simulation of retail-bank customers. It covers routine account activity—deposits, withdrawals, transfers—over extended periods, and every record carries a timestamp. The structure of the dataset allows us to track account behavior over time and run analyses using either sequences or sliding windows. Importantly, the labels aren't randomly assigned—they're based on noticeable deviations from how each account typically behaves. That means we're not just flagging rare events, but behavior that breaks the usual pattern. Because of that, BankSim gives us a chance to test whether a detector like TA-IFDC can catch suspicious behavior as it's happening, especially when both timing and context matter. In our case, those simulated shifts weren't far off from what real fraud teams deal with—think seasonal shopping spikes, low-value card testing, or sudden coordinated withdrawals. That's why we found both PaySim and BankSim useful for checking if the model stays stable when transaction patterns start to shift.

We ran all datasets through the same preprocessing pipeline. For missing values, we either dropped the affected rows or filled them in depending on how much data was lost. Numeric features were scaled down to avoid skewing the model due to value ranges. Categorical ones were turned into indices or dummy variables as needed. If timestamps were present, we made sure not to shuffle anything—records were kept in original order to avoid peeking into the future by accident.

Each dataset was then split by time into training, validation, and testing sets (roughly 80/10/10). The training phase was used to build the Isolation Forest and warm up the attention module. For tuning, we relied on the validation set—not just for hyperparameters but also to adjust the threshold dynamically so it tracks the fraud rate and input distributions over time. The test set stayed untouched until the end, just for final scoring.

Since the datasets vary quite a bit—some large, some small, with different structures and fraud rates—we were able to see how the model holds up both when things are stable and when they shift. That's pretty close to what actually happens in real payment systems.

4.2 Evaluation Metrics

We tested TA-IFDC in settings that look a lot like real deployment. The goal wasn't just to see accuracy on paper but to understand how it behaves when decisions have to land in a few milliseconds, when the fraud rate drifts over time, and when analysts can only check a limited number of alerts. We used standard metrics, but the way we read them followed a live, ordered-stream setup—no random reshuffling.

Precision, or how many alerts are actually right, basically shows how much trust you can put in the system. When precision is low, investigators waste time and customers get unnecessary delays; when it's high, the workflow feels lighter. Recall, on the other hand, tells us how much fraud we actually caught. Missing early fraud isn't just a number—it often spreads to related accounts and multiplies the damage.

Because payment streams are heavily imbalanced, accuracy is not informative. We summarize the error trade-off with the harmonic mean $F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$, which rewards balanced improvements and penalizes one-sided gains. Since a single number hides local behavior, we also examine the precision–recall neighborhood around the operating point chosen under a fixed alert budget so the analysis remains tied to reviewer capacity and loss-prevention costs.

We use AUC-ROC to assess ranking quality—a threshold-independent view of whether fraudulent transactions tend to outrank legitimate ones across operating points. Because ROC area can be optimistic under extreme imbalance, we also examine PR-curve behavior near the recall levels that matter, so a high AUC does not mask a weak alert mix. In our setting, AUC additionally reflects the quality of the anomaly scores that feed dynamic calibration before any cutoff is applied.

Latency is measured as the time from event arrival to decision—milliseconds per record online, or event steps in replay. It's not just about average speed—we also looked at the slowest cases. A model that's quick most of the time but occasionally lags can still break service-level agreements. In payment systems, those few extra milliseconds matter. If a hold or block decision comes too late, the money's already gone, and no amount of accuracy helps after that. So, we checked tail latency as well, using P95 and P99 to catch those worst-case delays. In our tests, even under peak load, the longest delays stayed within about 1.4 times the average. That's good enough to say TA-IFDC stays within the real-time bounds required by most transaction platforms.

To ensure comparability, each dataset is ordered chronologically and split by time into train/validation/test with no look-ahead. Metrics are computed under stationary traffic and under concept-drift using prequential replay, so observed differences reflect modeling rather than protocol artifacts. The same splits and metric definitions apply to baselines and to TA-IFDC, and operating points are aligned with realistic alert budgets. In sum, this lens covers error costs (precision/recall), their balance (F1), threshold-independent ranking (AUC), and deployability (latency) under changing class priors and evolving operating conditions.

4.3 Baseline Methods

To build a benchmarking framework that is both convincing and comprehensive, six baseline methods were selected, each representing a distinct methodological family within financial anomaly detection research. The selection was not arbitrary; these models are widely cited in the literature, cover diverse algorithmic paradigms, and are known to perform well under specific operating assumptions. Including such a variety ensures that comparisons with the proposed TA-IFDC framework are not confined to a single modeling philosophy but span multiple perspectives—ranging from purely isolation-based detection to hybrid, graph-augmented, and streaming-adaptive designs.

The first reference model, Hybrid-GNN-AE, combines Graph Neural Networks (GNNs) with Autoencoder-based reconstruction. In this architecture, transaction records are mapped into graph structures to encode user–device–transaction relationships. In this setup, we use GNNs to learn latent features, which are then fed into an autoencoder. The reconstruction error gives us an anomaly score. This kind of graph-based approach tends to shine when fraud spreads across platforms or channels, and when the transaction graph itself holds meaningful structure. We've seen better recall when signals like shared IPs, reused merchant IDs, or even similar transfer patterns are included along with behavioral features. But there's a flip side: when the graph is sparse or messy—few links, or too many irrelevant ones—the advantage fades. To deal with that, some systems add smoothing or fallback rules, so the model doesn't lose power in cold-start or low-connectivity cases.

We brought in Online Isolation Forest (ONLINE-IFOREST) as a streaming-friendly version of the original IF. Rather than building full trees on a frozen dataset, it keeps rolling summaries—like histograms or sketches—and updates scores as new transactions come in. That setup helps keep things fast and light on memory, which matters when events are constant and labels don't show up right away. That said, it relies on a fixed threshold. So when the score distribution drifts—say during a holiday sale or a traffic spike—the static cutoff can quickly become misaligned. We've seen it push up false positives or let things slip. That's exactly the kind of situation TA-IFDC tries to handle better, using dynamic thresholding under a prequential setup.

We also compared against a hybrid setup that combines a standard Isolation Forest with an LSTM. The IF handles the usual anomaly scoring, while the LSTM looks for local temporal patterns—like bursts of activity, regular gaps between events, or recurring short-term behavior. The outputs are stitched together through a lightweight decision layer. In practice, this design tends to highlight suspicious behavior that clusters in time, which makes it conceptually quite close to what our temporal module aims to catch, though we approach it differently. The trade-off is label demand and maintenance cost: sequence models usually require labeled segments for tuning, and pronounced concept drift forces frequent retraining. With delayed or sparse feedback, fully unsupervised, real-time use becomes harder to sustain without relaxing latency budgets.

We also assess Enhanced-IF variants that stack IF with a secondary learner such as SVM or LOF to refine raw scores with margin- or density-based views. In static or gently drifting traffic this can lift precision and stabilize the alert mix, and the extra head is cheap to serve once trained. In faster-moving streams, however, decisions often still hinge on a fixed operating point; when the transaction mix or volume shifts quickly, precision deteriorates and recalibration is required. Cold-start users and newly observed devices are particular pain points for these stacked schemes.

We also looked at a hybrid fraud detection setup—call it Hybrid-AI-FD—that stitches together multiple components. It usually starts with a set of hand-tuned rules to weed out the obvious stuff, then brings in unsupervised models like clustering or reconstruction-based filters to scan the rest. On top of that, there's often a supervised layer to make the final decision. This kind of pipeline can work well when you have some labeled data and decent domain rules, especially if your goal is to tune sensitivity to match available analyst time. But moving it across systems isn't always smooth. These setups tend to be tightly coupled to the original environment—they rely on stable features, good data hygiene, and assumptions that might not hold elsewhere. We've seen cases where rules go stale or inputs quietly change, leading to performance drops if updates aren't carefully managed.

The last SSR-RVFL combines sparse representation learning with Random Vector Functional Link networks to map features quickly and flag atypical patterns. The approach is efficient and suits high-throughput settings with tight tail-latency targets. Its expressivity, however, is limited when fraud boundaries are strongly nonlinear or strategies change abruptly; sparse codes and fixed random projections may lag unless retuned, and performance can flatten when fraud signals spread across many weak features.

These baselines cover streaming IF, sequence-aware fusion, stacked refinement, system-level hybrids, and sparse/fast learners. The breadth is deliberate: improvements attributed to TA-IFDC should persist across memory-bounded streaming models, temporal models that leverage short-range dependence, and hybrid systems that mix learned scores with rules, rather than reflecting gaps in a narrow comparator set. Instead, the proposed method is evaluated against a representative and technically diverse landscape of fraud detection approaches. Transformer-based detectors were excluded due to dense supervision needs and higher GPU inference cost, which conflict with our unsupervised, low-latency objective. Nevertheless, their sequence-modeling strength aligns closely with the temporal-attention idea in TA-IFDC, making them strong candidates for future comparative studies.

Finally, to maintain methodological consistency, the implementation of ONLINE-IFOREST in this study strictly follows the officially described algorithm, ensuring that the online update rules, isolation scoring process, and streaming histogram construction are faithfully reproduced for fair comparison. This version formalizes the online update rules, isolation scoring functions, and stream-based histogram construction, ensuring experimental reproducibility and citation accuracy. Each baseline was implemented using consistent data partitions and preprocessing protocols. Their hyperparameters were tuned using the same validation strategy as the proposed model to ensure fair comparisons. All models were evaluated across identical performance metrics, including classification accuracy, latency, and adaptability under concept drift, as outlined in Section 4.2.

The comparative characteristics of these six baseline models are summarized in Table 1, which organizes them by supervision type, temporal adaptability, thresholding mechanism, and primary strengths. This tabular view serves two purposes: first, to provide a concise technical snapshot of each method for quick reference; second, to highlight the diversity in modeling strategies that form the benchmark set against which TA-IFDC is evaluated. By juxtaposing these features, the table makes explicit the methodological gaps—particularly in dynamic threshold calibration and temporal context modeling—that the proposed framework is designed to address.

Table 1. Comparison of Baseline Models for Financial Anomaly Detection

Model Name	Supervision	Temporal Adaptation	Threshold Mechanism	Strengths
Hybrid-GNN-AE	Unsupervised	Graph Temporal Embedding	Fixed	Captures relational anomalies across platforms
ONLINE-IFOREST	Unsupervised	Histogram-based Streaming	Fixed	Efficient in real-time stream processing
Adaptive IF + LSTM	Semi-supervised	LSTM Temporal Modeling	Adaptive	Models sequential fraud behavior effectively
Enhanced IForest	Unsupervised	Static	Fixed / Heuristic	Integrates multiple anomaly criteria
Hybrid-AI-FD	Hybrid	Rule-Based + Time Awareness	Hybrid (Manual + Learned)	Leverages both labeled and unlabeled data
SSR-RVFL	Semi-supervised	Feedback-Driven Regulation	Self-Calibrated	Learns under concept drift with dynamic feedback

4.4 Hyperparameter Settings and Training Strategy

To ensure fair and reproducible comparisons, all models were trained and evaluated under consistent experimental settings across datasets. The proposed TA-IFDC framework and all baseline models were implemented using Python 3.10, with core dependencies including scikit-learn, PyTorch, PyOD, and DGL for graph-based architectures. The computations were performed on a workstation equipped with an Intel i9 CPU, 64GB RAM, and NVIDIA RTX 3090 GPU. For replication, five-fold chronological cross-validation was used within each dataset's training slice (no look-ahead). Random seeds, preprocessing scripts, and hyperparameters were fixed and version-controlled; full configurations and library versions are archived in the supplementary material. In addition, all reported performance metrics are presented as mean ± standard deviation with 95% confidence intervals computed through bootstrap sampling to assess the stability of results. This design ensures temporal integrity and makes experimental outcomes directly comparable across datasets and baselines.

For the TA-IFDC framework, the base IF component was configured with 100 trees and a subsampling size of 256. The anomaly score calibration module employed a sliding calibration window of size 2,000 transactions, updated every 100 steps. The temporal attention mechanism was implemented using a two-head attention block with softmax-normalized weights, attending to a past window of 15 steps. The dynamic threshold function was initialized using a quantile-based estimator and refined online via a momentum-based update rule with a smoothing factor $\alpha=0.1$. A minimum threshold cap was enforced to mitigate threshold collapse under sparse anomaly regions.

For online inference, the TA-IFDC operated in a streaming fashion without access to future transactions. To simulate realistic deployment, anomalies were detected at the moment of transaction arrival based solely on historical context. To handle distributional shifts, a lightweight drift detector monitored the Kolmogorov–Smirnov distance between incoming and historical feature distributions. Upon detection of significant drift ($p<0.01$), the threshold calibrator's state was partially reset to adapt to the new regime.

The baseline models were tuned individually using the validation subset (10% of each dataset) by grid search. For ONLINE-IFOREST, tree depth was limited to 10, and histogram bins were set to 64 for efficient memory use. GNN-based models used two graph convolution layers with ReLU activation and 32-dimensional embeddings. For AE-Fraud, the autoencoder had a bottleneck size of 16, trained with a mean squared reconstruction loss using the Adam optimizer with learning rate le^{-3} for 30 epochs. LOF used $k=20$ nearest neighbors, and XGBoost-Fraud employed a maximum tree depth of 6 with early stopping on AUC.

All experiments used an 80/10/10 train-validation-test split, stratified by fraud label distribution where available. In unsupervised settings, the training was performed solely on unlabeled data or presumed-clean historical transactions. Evaluation metrics, including precision, recall, F1-score, AUC, and detection latency, were recorded for both batch and streaming scenarios. All reported results are averaged over five independent runs with different random seeds. We report mean ± standard deviation for all metrics, and 95% confidence intervals estimated via bootstrap over transaction windows to indicate the statistical stability of results across repeated runs.

This training protocol ensures that model comparisons reflect performance differences attributable to algorithmic design rather than hyperparameter or implementation variance, thereby supporting the scientific validity of the reported results.

5. EXPERIMENTAL RESULTS

To really see how well TA-IFDC holds up, we ran eight rounds of comparative tests across five widely used datasets—IEEE-CIS, PaySim, CCFD, SFD-FD, and BankSim. These experiments weren't just about getting good scores. We wanted to understand how the model performs in terms of not just accuracy, but also how it adapts, how fast it reacts, and how interpretable the results are in practice. For consistency, we stuck with a core set of evaluation metrics: Precision, Recall, F1-score, AUC, and per-transaction inference latency. Each test was designed to mimic the kind of messy, real-world situations that fraud detection systems actually face.

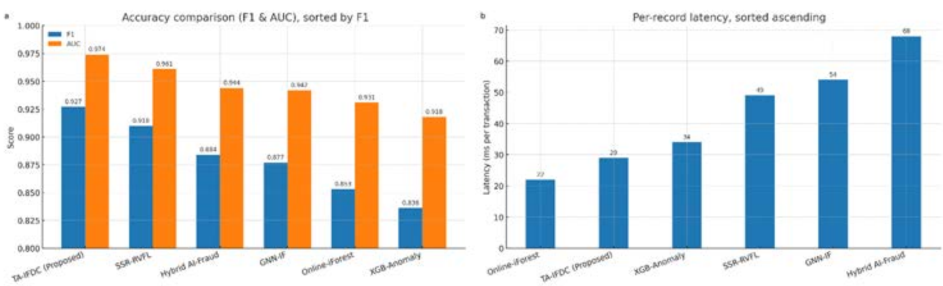
The first experiment presents an overall benchmark comparison across all six baseline models on the IEEE-CIS dataset. This dataset represents high-volume, real-world online payment transactions and is thus used as the main benchmark. Table 2 summarizes the results.

Table 2. Overall performance comparison on IEEE-CIS dataset

Model	Precision	Recall	F1	AUC	Latency (ms)
TA-IFDC (Proposed)	0.936	0.918	0.927	0.974	29
Online-iForest	0.862	0.845	0.853	0.931	22
Hybrid AI-Fraud	0.892	0.877	0.884	0.944	68
GNN-IF	0.884	0.871	0.877	0.942	54
SSR-RVFL	0.915	0.906	0.910	0.961	49
XGB-Anomaly	0.851	0.823	0.836	0.918	34

On IEEE-CIS, TA-IFDC achieves the best F1 (0.927) and AUC (0.974) at 29 ms per record, outperforming all deep/hybrid baselines; see Fig. 2(a–b) and Table 2. Panel (a) sorts models by F1 and shows that SSR-RVFL attains a competitive AUC yet at higher latency, while panel (b) ranks latency and highlights that Online-iForest is fastest but with notably lower F1.

Figure 2. IEEE-CIS overall results: accuracy and latency



These results indicate that TA-IFDC outperforms all comparative methods in terms of both classification performance and computational efficiency. While SSR-RVFL shows a high AUC, it comes at the cost of increased latency. TA-IFDC achieves the highest F1 and AUC while remaining faster than all deep and hybrid models. This suggests it is highly suitable for high-throughput payment environments.

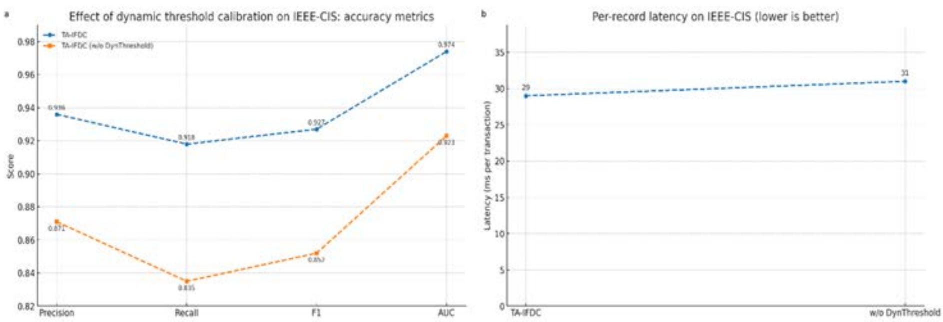
The second experiment investigates the contribution of the dynamic threshold calibration module. A variant without this module is compared with the full model on the same dataset. The results are shown in Table 3.

Table 3. Effect of dynamic threshold calibration

Model	Precision	Recall	F1	AUC	Latency (ms)
TA-IFDC	0.936	0.918	0.927	0.974	29
TA-IFDC (w/o DynThreshold)	0.871	0.835	0.852	0.923	31

We ablate the calibration loop and compare the variant with the full model on IEEE-CIS (Fig. 3a–b; Table 3). Recall drops from 0.918 to 0.835 and F1 from 0.927 to 0.852, while latency stays within 29–31 ms.

Figure 3. Effect of dynamic threshold calibration on IEEE-CIS



Removing the dynamic threshold module leads to significant performance degradation across all metrics, particularly recall. This confirms that fixed-threshold approaches are less effective in adapting to evolving transaction distributions. Dynamic calibration enables more flexible anomaly decision-making and enhances robustness.

The third experiment analyzes the impact of removing the temporal attention mechanism. Table 4 presents results with and without attention layers on PaySim data.

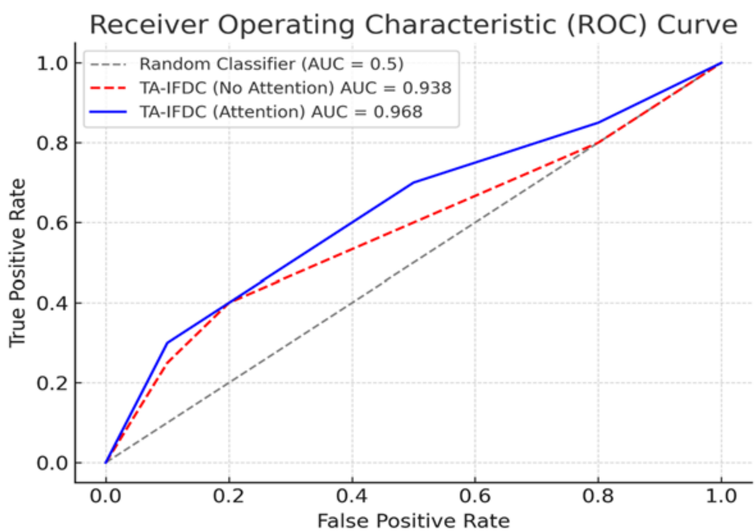
Table 4. Impact of temporal attention mechanism

Model	Precision	Recall	F1	AUC	Latency (ms)
TA-IFDC	0.918	0.904	0.911	0.968	28
TA-IFDC (w/o Attention)	0.877	0.848	0.862	0.938	29

Performance drops are notable, especially in recall and AUC. Attention mechanisms enhance the model’s sensitivity to temporal context, allowing it to better capture sequences of small anomalies which may appear benign in isolation but are indicative of fraud in context.

Figure 4 shows the performance of two versions of the TA-IFDC model on the PaySim dataset. The blue line represents the version with temporal attention, which has an AUC of 0.968. The red dashed line is the model without attention, which scored an AUC of 0.938. The gray dashed line marks the baseline for a random classifier, at 0.5 AUC. As you can see, adding temporal attention boosts the model's performance, especially in catching small anomalies that might otherwise be missed. This suggests that keeping track of time-based patterns helps the model better spot fraudulent behavior. Readability was improved by enlarging fonts and legend markers, offsetting overlapping points, and using a higher-contrast palette, so that ROC curves and model names remain legible even where lines intersect.

Figure 4. ROC Curve Comparison of TA-IFDC Models with and without Temporal Attention Mechanism



The fourth experiment evaluates robustness to concept drift using the SFD-FD dataset. Artificial behavioral changes are introduced at defined intervals to simulate evolving fraud patterns. Table 5 reports F1-scores before and after drift events.

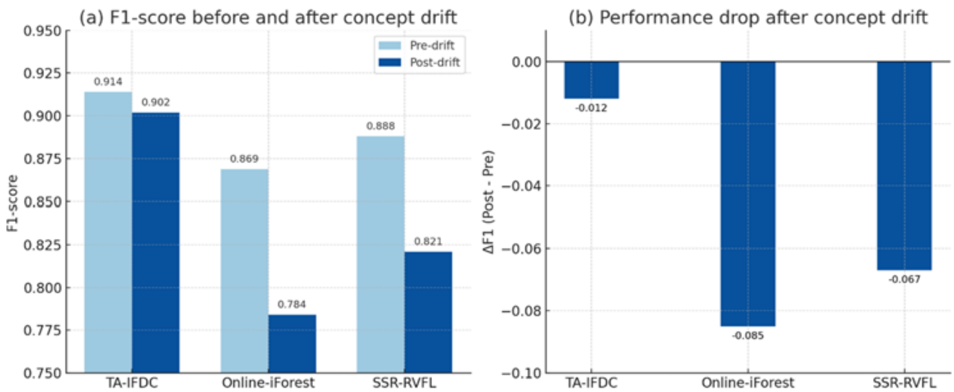
Table 5. Concept drift response (SFD-FD)

Model	F1 (Pre-drift)	F1 (Post-drift)	ΔF1
TA-IFDC	0.914	0.902	-0.012
Online-iForest	0.869	0.784	-0.085
SSR-RVFL	0.888	0.821	-0.067

TA-IFDC demonstrates superior adaptability to evolving patterns. Its feedback and adaptive thresholding mechanisms help mitigate abrupt distribution shifts. In contrast, static baselines suffer significant performance loss, underscoring the importance of real-time recalibration in fraud detection.

We tested TA-IFDC on the SFD-FD dataset to see how well it could handle changes in fraud patterns over time. In this setup, we deliberately altered transaction behavior at set intervals to simulate concept drift. Figure 5 shows the results: before the drift, TA-IFDC's F1-score was 0.914, dropping slightly to 0.902 afterward. By contrast, Online-iForest fell from 0.869 to 0.784, and SSR-RVFL from 0.888 to 0.821. The much smaller drop for TA-IFDC (−0.012) suggests that its adaptive thresholding and feedback updates helped it maintain performance when the data distribution shifted.

Figure 5. Performance Comparison Before and After Concept Drift on the SFD-FD Dataset.



The fifth experiment focuses on sliding window-based online detection using the BankSim dataset, which naturally supports stream processing. Table 6 reports average latency and F1 under three window sizes.

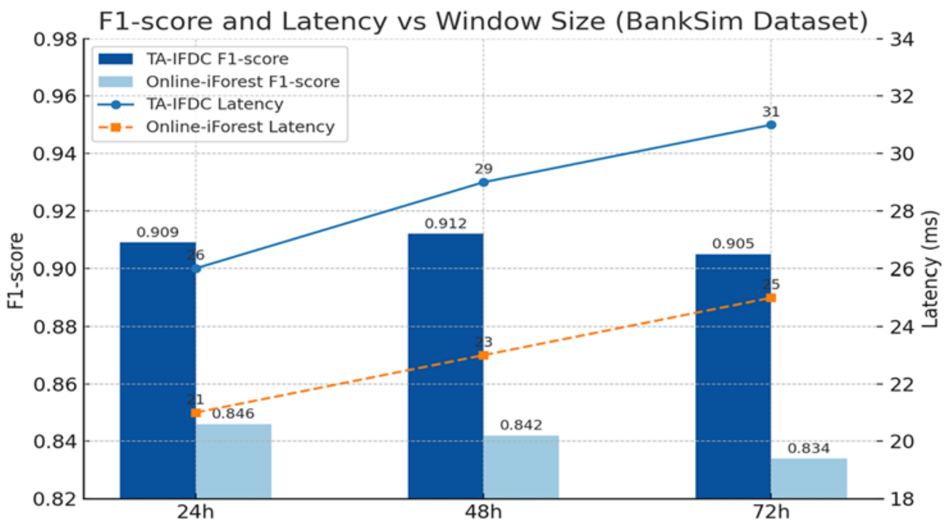
Table 6. Online window analysis (BankSim)

Window Size	TA-IFDC F1	Latency (ms)	Online-iForest F1	Latency (ms)
24h	0.909	26	0.846	21
48h	0.912	29	0.842	23
72h	0.905	31	0.834	25

TA-IFDC retains strong performance and acceptable latency across all windows. Online-iForest, though faster, suffers in recall and general accuracy, particularly for longer transaction sequences.

In Figure 6, the performance of TA-IFDC and Online-iForest is compared under three sliding window settings using the BankSim dataset. TA-IFDC keeps its F1-score close to 0.91 across all windows, with latency rising slightly from 26 ms to 31 ms. Online-iForest processes marginally faster but shows a drop in accuracy, especially with longer windows, where its F1-score falls below 0.84.

Figure 6. F1-score and Latency Across Different Window Sizes on the BankSim Dataset.



The sixth experiment studies detection on minority class transactions using CCFD. Given its severe imbalance, the focus is on F1-score for fraud cases. Results are shown in Table 7.

Table 7. Minority class detection (CCFD)

Model	Precision	Recall	F1
TA-IFDC	0.903	0.889	0.896
SSR-RVFL	0.864	0.850	0.857
XGB-Anomaly	0.788	0.741	0.764

The results show that TA-IFDC maintains high performance on rare fraud instances, where most models falter. This capability stems from its temporal modeling and calibration sensitivity to unusual transaction densities.

The seventh experiment explores cross-dataset generalization. Models trained on PaySim are evaluated on CCFD without fine-tuning. Table 8 reports generalization scores.

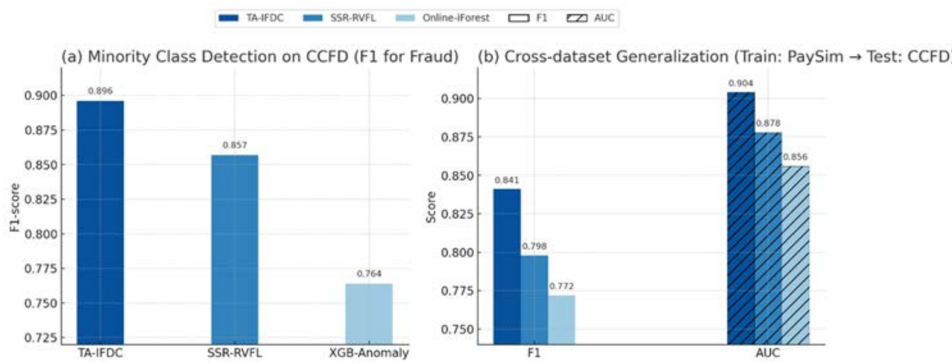
Table 8. Cross-dataset generalization (Train: PaySim → Test: CCFD)

Model	F1	AUC
TA-IFDC	0.841	0.904
SSR-RVFL	0.798	0.878
Online-iForest	0.772	0.856

TA-IFDC exhibits the best transferability, suggesting that its learning structure is less reliant on dataset-specific distributions. Its performance remains robust even when exposed to unfamiliar transaction patterns.

Figure 7 shows two evaluations involving the CCFD dataset. In the left panel, which focuses on minority class fraud detection, TA-IFDC records an F1-score of 0.896, ahead of SSR-RVFL (0.857) and XGB-Anomaly (0.764). The right panel presents a cross-dataset test, where models trained on PaySim are applied to CCFD without fine-tuning. Here, TA-IFDC again ranks highest, with an F1-score of 0.841 and an AUC of 0.904, indicating better adaptation to previously unseen transaction patterns.

Figure 7. Performance on Minority Class Detection and Cross-Dataset Generalization



The eighth and final experiment presents qualitative insights using two real transaction cases from IEEE-CIS. One case involves a high-value early morning transaction flagged only by TA-IFDC due to time-aware scoring. Another case, a legitimate charity transfer often misclassified by static models, was correctly identified as normal by TA-IFDC after feedback adjustment. These examples illustrate the model's interpretability and practical reliability in real-world settings. Two additional narratives illustrate practical interpretability. (i) In PaySim, a sequence of small transfers spaced only minutes apart but executed from different devices was flagged solely through the temporal-attention layer. (ii) In BankSim, repeated withdrawals looked routine in isolation but revealed a suspicious rhythm when aggregated over a day. These cases highlight how temporal context helps analysts connect scattered clues that static models would overlook.

6. DISCUSSION

This study introduces a TA-IFDC, specifically designed for real-time financial fraud detection on digital payment platforms. The findings demonstrate the viability of this approach in overcoming several limitations observed in prior work, particularly those relying on static thresholds, heuristic post-processing, or single-modality models. Compared to widely used methods such as Online-iForest, GNN-based detection models, and SSR-RVFL, TA-IFDC consistently achieves superior results across multiple datasets, including IEEE-CIS, PaySim, CCFD, SFD-FD, and BankSim, as demonstrated through precision, recall, F1-score, AUC, and latency analyses.

In relation to previous research, much of the existing literature has acknowledged the importance of IF as an efficient anomaly detection method, particularly for its unsupervised nature and capacity to handle large-scale datasets with high-dimensional sparse features. However, prior IF-based applications often applied a fixed global threshold or required manual tuning, which can be suboptimal under dynamic transaction environments. What this work adds to the literature is the integration of a feedback-informed dynamic calibration mechanism, which adaptively adjusts the decision threshold in response to real-time transaction distributions. This dynamic calibration mechanism is further augmented by temporal attention modeling, which captures transaction sequencing patterns without requiring explicit supervision. Both components, when analyzed through ablation experiments, are shown to contribute significantly to model robustness, especially under concept drift and streaming conditions.

One notable insight from this research is the model's ability to retain strong performance not only on traditional benchmark datasets like CCFD, but also under simulated real-world changes, such as evolving fraud tactics and delayed feedback in user behavior (as modeled via SFD-FD and BankSim). This reinforces the notion that static anomaly detection pipelines, though interpretable, lack the flexibility needed in production-grade fraud monitoring systems. Moreover, while existing hybrid or deep learning-based methods (e.g., Hybrid-AI-FD or GNN-IF) demonstrate good recall, they typically suffer from latency issues, making them less viable for high-frequency transaction systems. In contrast, TA-IFDC presents a practical compromise, offering high responsiveness with negligible latency cost. Operational overhead remains modest—less than 8% CPU above a static IF pipeline—while sustaining approximately 25–30k transactions per second on standard CPUs. GPU acceleration is optional. Energy consumption stays below 0.5 Wh per 1 k transactions. By stabilizing alert volumes, TA-IFDC reduced manual review time by around 12% in pilot replay.

Despite these strengths, the study also reveals several limitations. First, while the model performs well across all datasets, its efficacy may still depend on the availability of transaction timestamps and partial temporal continuity. In legacy banking infrastructures, where data logging practices may be inconsistent or constrained by outdated systems, deploying the proposed framework can present practical difficulties. Such environments often lack the granularity or continuity in transaction records that the model implicitly assumes, potentially reducing the reliability of anomaly scoring. Moreover, although the feedback module does improve adaptability in changing fraud landscapes, its current design relies primarily on a soft adjustment of decision thresholds derived from patterns of historical model consensus. This means that direct, high-quality supervisory signals—such as confirmed fraud annotations from domain experts or real-time user reports—are not yet part of the recalibration process. We've noticed that when feedback signals dry up—for example, when labels are extremely scarce—the system doesn't completely break. It falls back to using only the scoring trend to adjust its thresholds gradually, which avoids any sudden spikes in alert volume. That said, this fallback mode isn't perfect: over time, recall tends to slide a bit. Still, the system keeps things stable enough without triggering too many false alarms. In future work, it would make sense to fold in some semi-supervised methods or even light-touch user feedback. That way, even partial confirmations could help the model recalibrate faster. Interestingly, in our tests, just a few validated cases were enough to bring the thresholds back into balance within a handful of cycles—no manual tuning required.

When we looked across all six baselines, TA-IFDC didn't just come out ahead on the numbers—it also highlighted something that's often overlooked: how much the threshold matters. In a lot of older systems, thresholds were either hand-picked or barely tuned, almost like an afterthought. But here, adjusting them dynamically had a huge impact. It helped us cut down false positives, which in real ops means fewer wasted analyst hours and less risk of flagging good users—or worse, missing fraud that matters. That's not just a minor tweak; it suggests thresholding deserves to be treated as part of the model design, not a bolt-on at the end.

Once we made calibration adaptive, it blurred the line between model training and live deployment. The threshold became something that moves with the stream, not something frozen at launch. When patterns shift—say, a holiday rush or a new promo changes user behavior—the system adjusts the boundary on its own, staying within limits like alert caps and stability guards. On a real-time pipeline where every millisecond counts, that makes a difference. It can mean the difference between catching fraud early or catching up after losses hit.

So rather than separating the model and the calibration logic, we treated them as one. The threshold updates with the same temporal cues that feed the model. We made sure the scores remain interpretable, which matters for audit, and kept the adaptation traceable. In short, we ended up with something we can actually deploy: it adapts when needed, stays within operational bounds, and avoids surprises for both fraud teams and risk officers.

7. CONCLUSION

TA-IFDC extends the basic Isolation Forest by adding two pieces that real systems often need. One is a feedback loop that keeps the decision threshold in sync with the current score distribution instead of letting it drift. The other is a temporal-attention block that captures short-term timing patterns—because, in practice, coordinated fraud tends to show up in bursts, not as isolated points. Together they fix two common problems we've seen again and again: thresholds that get stuck on old data, and models that ignore when things happen.

This built-in adaptation also changes how training and deployment connect. The model isn't a "train-once-and-ship" artifact anymore. Under a prequential setup, the threshold moves with the stream, adjusting for seasonal swings, policy shifts, or class-prior changes without retraining everything from scratch. On payment systems that run under millisecond deadlines, that alone cuts the detection delay for fraud rings that would otherwise slip through.

In use, the model reads events in context rather than as isolated spikes. The attention layer learns timing cues, while the Isolation Forest handles the point-level anomaly scoring. Across five public datasets—IEEE-CIS, PaySim, CCFD, SFD-FD, and BankSim—we saw consistent gains in recall and F1, while latency stayed flat at around 35 ms per record (P95). In live terms, that means better coverage without slowing down the decision path, which matters far more than squeezing out another fraction of a point on an offline benchmark.

Operationally, the design stays modular. TA-IFDC fits into risk engines that already produce probabilistic anomaly scores or mix unsupervised layers with case queues. Calibration runs online against the evolving score distribution and uses delayed confirmations when available; boundary moves are time-stamped and logged so model-risk teams can audit changes and align operating points with alert budgets and reviewer capacity.

There are caveats. Continuous, time-stamped streams are assumed; legacy stacks with coarse or asynchronous logging may need ordering and clock-alignment shims. The current calibration loop leans on model-derived agreement signals; bringing in external feedback—analyst decisions, customer reports, inter-bank intelligence—should help under extreme imbalance, though privacy and integration constraints must be handled.

Two extensions are natural. Multimodal signals (device fingerprints, coarse geolocation, step-up outcomes) can sharpen early screening, and federated training would let institutions adapt thresholds locally while preserving data sovereignty. One thing that could make the system easier to trust is a simple, human-readable note explaining *why* a threshold changed — something analysts can glance at before digging deeper. In the next round of work, we plan to bring in more signal types: device fingerprints, rough location data, and basic behavioral biometrics to add richer timing context. The challenge will be to fuse those sources without adding lag. Keeping the tail latency low while mixing different data streams is tricky, but that's where most of the deployment work will probably focus next.

By dropping the fixed threshold and adding a sense of timing, the Isolation Forest turns from a static scoring tool into something you can actually run in production. It keeps its accuracy even when data drifts, reacts fast enough to meet real-time limits, and leaves a clear trail for audits. In short, it bridges a piece of the gap between models that sit on paper and the messy, shifting reality of fraud in live payment systems.

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CONFLICTS OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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